

# GOTO **COPENHAGEN 2025**

September 29 - October 3 • TAP1, Copenhagen • [gotocph.com](https://gotocph.com)

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# Leading ~~3000~~ 5000 developers into the next era of Software Engineering

**Stuart Greenlees**

Senior Director of  
Technology, Liberty IT

**Garth Gilmour**

Technical Learning  
Consultant, Liberty IT

*Everything we wish we  
had known but found  
out the hard way.*



**Stuart**

MA in Psychology

MSc from QUB

Trains in Powerlifting

Employed at Liberty IT since 2004

Senior Director of Technology

Chunky Peanut Butter



**Garth**

BA in Psychology

MSc from UUJ

Trains in Weightlifting

Contracting at Liberty IT since 2003

Technical Learning Consultant

Smooth Peanut Butter





GenAI adoption was a Roller Coaster.



# Deploying at scale in Liberty Mutual

Tens of millions of customers in 29 countries and economies

50,000 employees worldwide, including 5,000+ technologists

Founded in 1912 – more than 110 years of innovation

Investing heavily in AI, Cloud, and Digital Transformation

**“To help people embrace today and  
confidently pursue tomorrow”**





“We’re going to get our knees skinned”



1

# Learning never stops

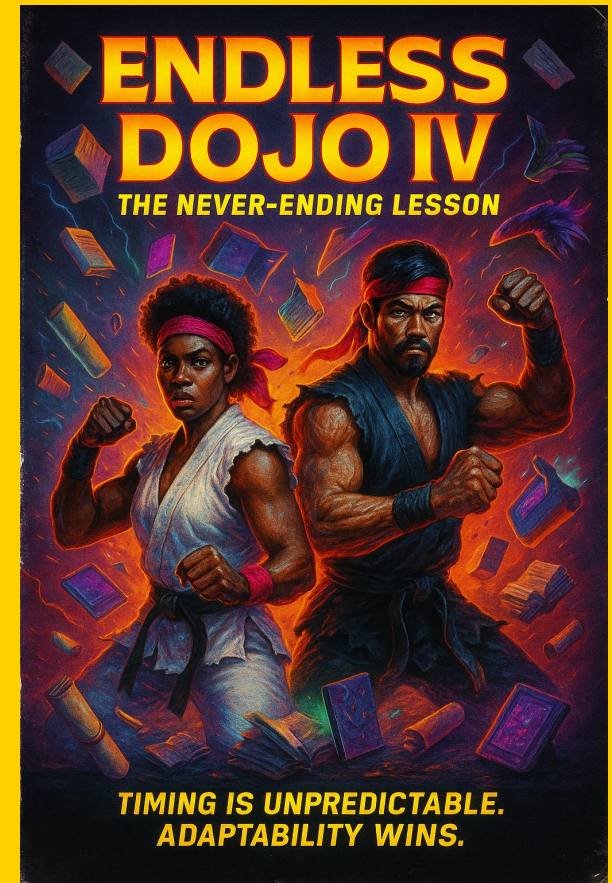


Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is "Endless Dojo IV: The Never-Ending Lesson". Show two diverse martial arts heroes who are software architects (different genders, ethnicities, and body types — stylized, generic, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: [Two martial artists surrounded by scrolls, books, and glowing tech devices raining from the sky.]. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: "[Timing is unpredictable. Adaptability wins.]"."





# Right tech, wrong time

## Thanksgiving 2022 we researched GenAI

Along with two other emerging technologies  
Our verdict – “worthy of future investigation”

## ChatGPT dropped *the next week*

We left the party just as it got good



Image generated by ChatGPT using the prompt: "Photorealistic close-up of an adult baseball batter mid-swing clearly missing the ball by inches. White jersey with thin blue pinstripes and cobalt-blue cap, no logos. Wooden bat; subtle motion blur on bat/hands. Baseball passes just in front of the batter's left cheek (no impact), eyes squeezed shut. Shot in a ballpark with blue bokeh crowd; shallow depth of field, natural daylight. Crisp focus on face and ball; high detail (skin texture, fabric weave).



# NOVEMBER 2022

| SUN | MON | TUE | WED | THU | FRI | SAT |
|-----|-----|-----|-----|-----|-----|-----|
| 30  | 31  | 1   | 2   | 3   | 4   | 5   |
| 6   | 7   | 8   | 9   | 10  | 11  | 12  |
| 13  | 14  | 15  | 16  | 17  | 18  | 19  |
| 20  | 21  | 22  |     | 24  | 25  | 26  |
| 27  | 28  |     |     |     | 2   | 3   |

**ChatGPT  
Launches**



# We bounced back!

**Used our research to brief Leadership fast**

**Leveraged our partnership with Azure to:**

- Quickly get access to the Open AI models

- Build an internal ChatGPT clone 'LibertyGPT'

- Stand up an MVP AI Platform

**Maintained control of our data**

- By blocking access to ChatGPT

- By offering the capability via LibertyGPT





# Lessons to learn

## Do the homework

Even if it's not the "winner"  
It builds future agility

## Be ready to pivot

Timing is unpredictable  
Adaptability wins!

## Leverage partnerships

Use existing vendors for speed and safety

## Keep guardrails in place

Don't compromise in a rush to deliver



2

# Shipping isn't adoption

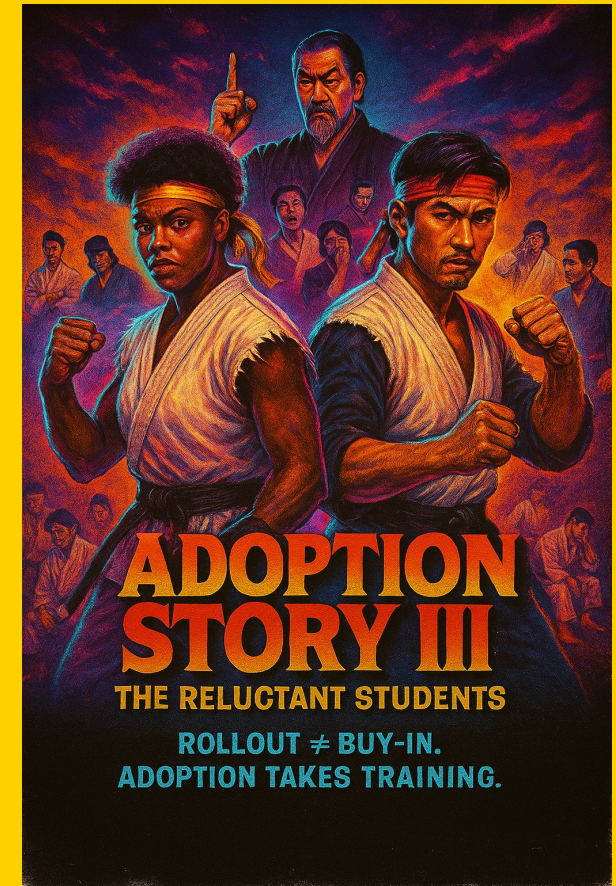


Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is "Adoption Story III: The Reluctant Students". Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artist sensei teaching students; half are distracted, asleep, or walking away. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: "Rollout ≠ buy-in. Adoption takes training.""





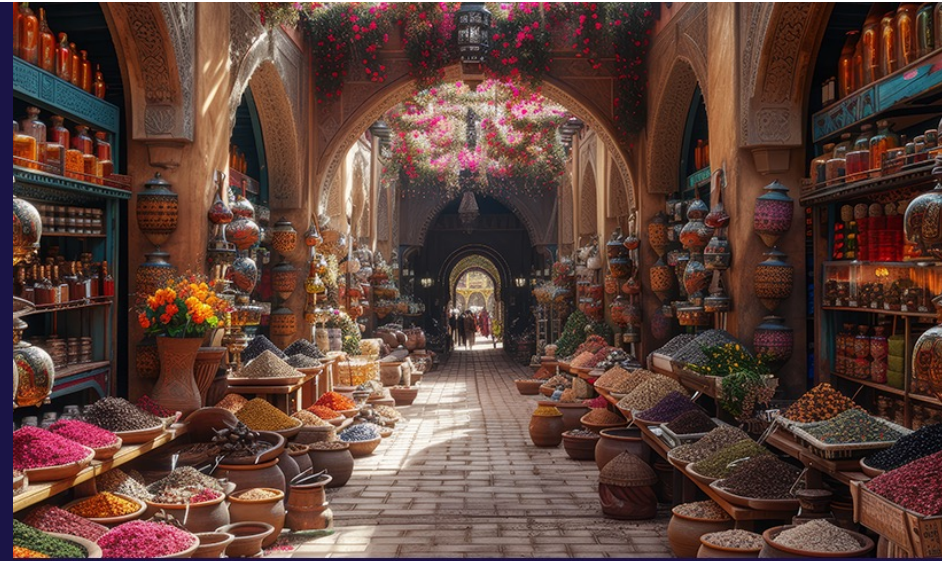
# Goal: Universal adoption

## Our dream is the Bazaar

LibertyGPT for all employees  
GitHub CoPilot for all coders  
Role-specific tools as needed

## We want to avoid the Cathedral

GenAI restricted to specialists





# We started strong!

## Initial progress was very fast

LibertyGPT is available in 70% of Liberty Mutual

## Rolled out GitHub Copilot org-wide

60% adoption on release in Liberty IT

Rose to 90% adoption after the first year

Started standardising instruction files



# Problem 1: LibertyGPT adoption is hard

**That last 30% is intractable**

**Models aren't always available**

Storage and Egress may conflict with Data  
Sovereignty Regulations e.g. GPT 5.0 isn't  
available in India

**Plus LLMs are often underused**

Many folks view them as a better Google  
Not as the Thought Partner it could be



# Problem 2: GitHub Copilot adoption is hard

## We don't have telemetry on GitHub Copilot

Hard to tell how heavily it is used

## Early indications were that usage was basic

Advanced features not being used effectively

## Unsurprising as Copilot is a moving target

Github CoPilot? Agent Mode?? Coding Agent???

Vibe coding became a methodology overnight

Spec-Driven Development (SDD) is the new shiny





# Solution: Inspire and educate

## Prompt-a-thons and unconferences

Kudos to everyone who volunteered!

## Inspirational sessions on Copilot

Prompting in Copilot (Wordle Helper)

Vibing with the Coding Agent (Wordle)

Instructions files (Command Line Utility)

## A separate GenAI Learning Mission

Made up of 8 separate educational streams

Architects, Learning & Development, Talent & Comms, and outside providers

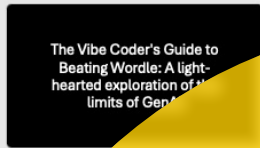




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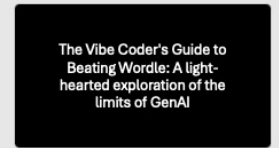
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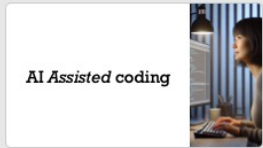
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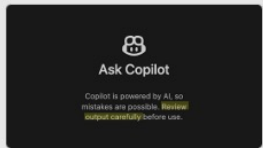
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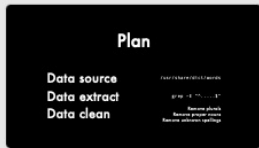
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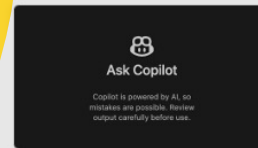
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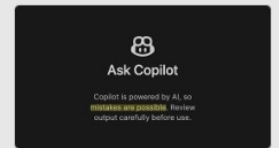
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# PROMPTING









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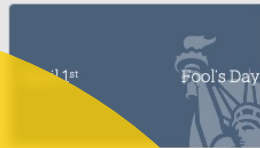
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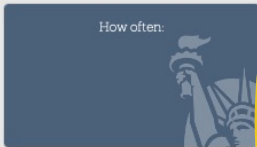
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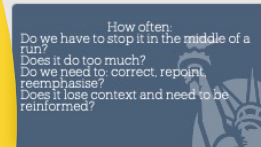
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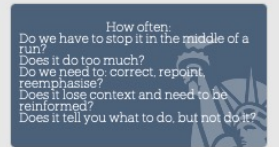
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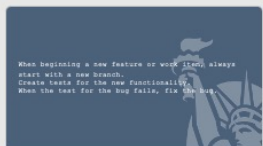
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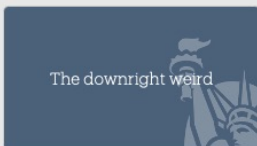
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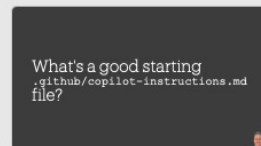
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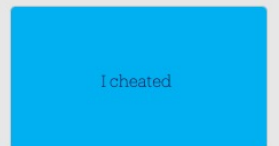
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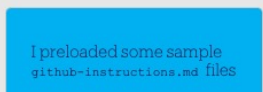
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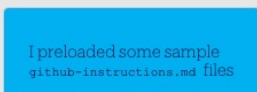
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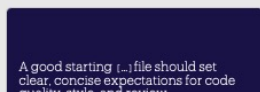
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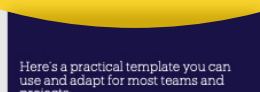
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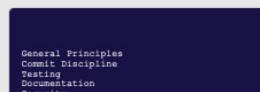
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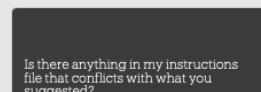
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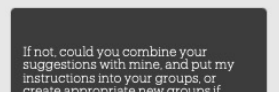
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# CONTROLLING



### Instructions File

If the user message starts with "Yo Fam", you start and end a response with a 2-3 sentence Gen-Alpha remark.

### Prompt

Yo fam, review this code

Yo fam, this terminal presentation tool is absolutely fire! Let me break down this sick codebase for you, no cap! 🔥



# Our GenAI Learning Mission

|                                       | Jun | Jul | Aug | Sept | Oct | Nov | Dec |
|---------------------------------------|-----|-----|-----|------|-----|-----|-----|
| Gen AI Literacy                       | →   |     |     |      |     |     |     |
| Tools & Ways of Working               | →   |     |     |      |     |     |     |
| Evolving Gen AI Skills                | →   |     |     |      |     |     |     |
| Including AI in Apps & Best Practices | →   |     |     |      |     |     |     |
| People Leader Journey                 | →   |     |     |      |     |     |     |
| Monthly open “office hours”           | →   |     |     |      |     |     |     |





# Lessons to learn

## **Adoption can be asymptotic**

The closer to 100% the harder it becomes  
All kinds of localization issues can occur

## **Adoption isn't effective use**

Measure real usage, not just installation

## **Effective use isn't fluency**

Inspire and educate your engineers

## **Fluency isn't efficiency**

Standardize across your teams



3

# Models aren't equal

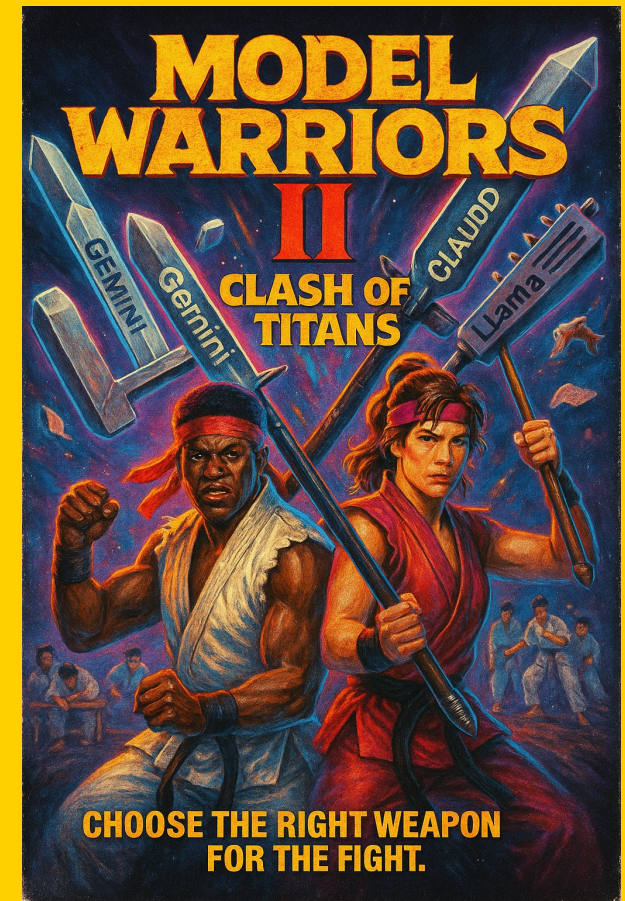


Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is 'Model Warriors II: Clash of the Titans'. Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artists wielding oversized weapons labeled OpenAI, Gemini, Claude, Llama — some sleek, some clumsy.. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: 'Choose the right weapon for the fight.'"



# Problem: Assuming models are fungible

**We assumed models were interchangeable**

Differentiated only by price and availability

**In reality, each behaves differently**

**We used stylometric analysis on content generated from technical material**

Gemini simplifies terminology

OpenAI keeps terminology

**We also have seen that some models are better with images, others with code**





# Solution: Support a plurality of models

We have a portfolio of models from leading AI providers and open-source communities.

We optimize for performance and cost

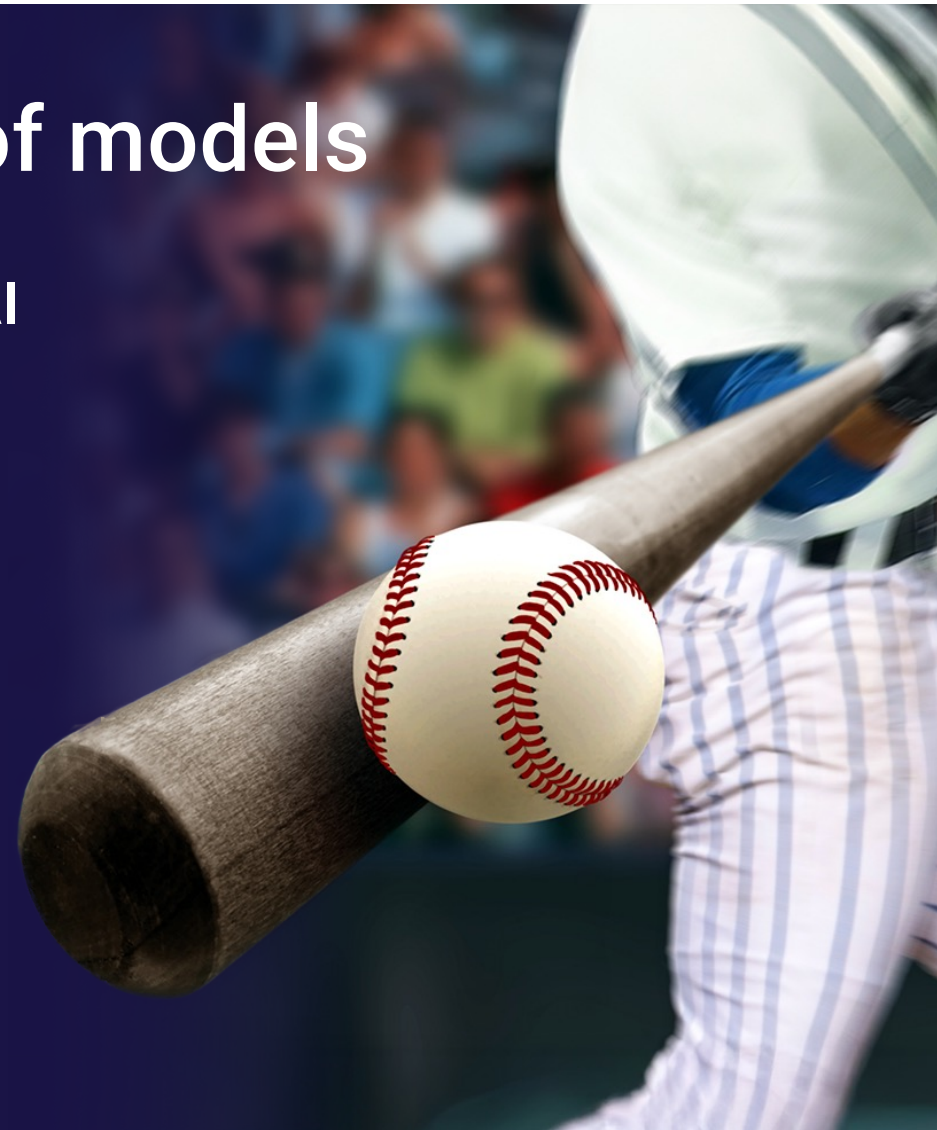
The “latest and greatest” is often the most expensive - and not always necessary.

The latest frontier model may also be slower.

**We always keep an eye on pricing**

Models become cheaper as new ones appear

Sometime the drop in price can be dramatic



# Lessons to learn

## Pick the right tool for the job

Choose based on context, not hype

## Balance cost against capability

Know how models work in your domain  
Test against representative workloads

## Don't assume interchangeability

Models differ in accuracy, latency, and style

## Continuously evaluate

Benchmark, measure, and revisit  
The model landscape shifts fast



4

# Scaling is a beast

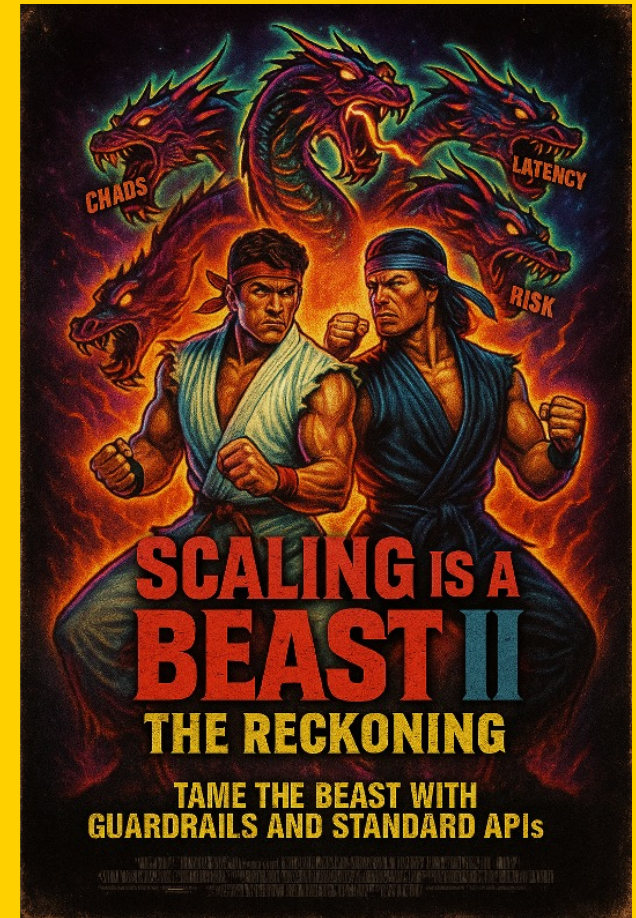


Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is 'Title: RAGged Edge II: The Complexity-Berg'. Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artists climbing a giant iceberg where only the tip is visible, the massive underside labeled Chunking, Embeddings, Eval Pipelines.. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: 'Off-the-shelf gets you 60%. Mastery takes sweat and scars.'"





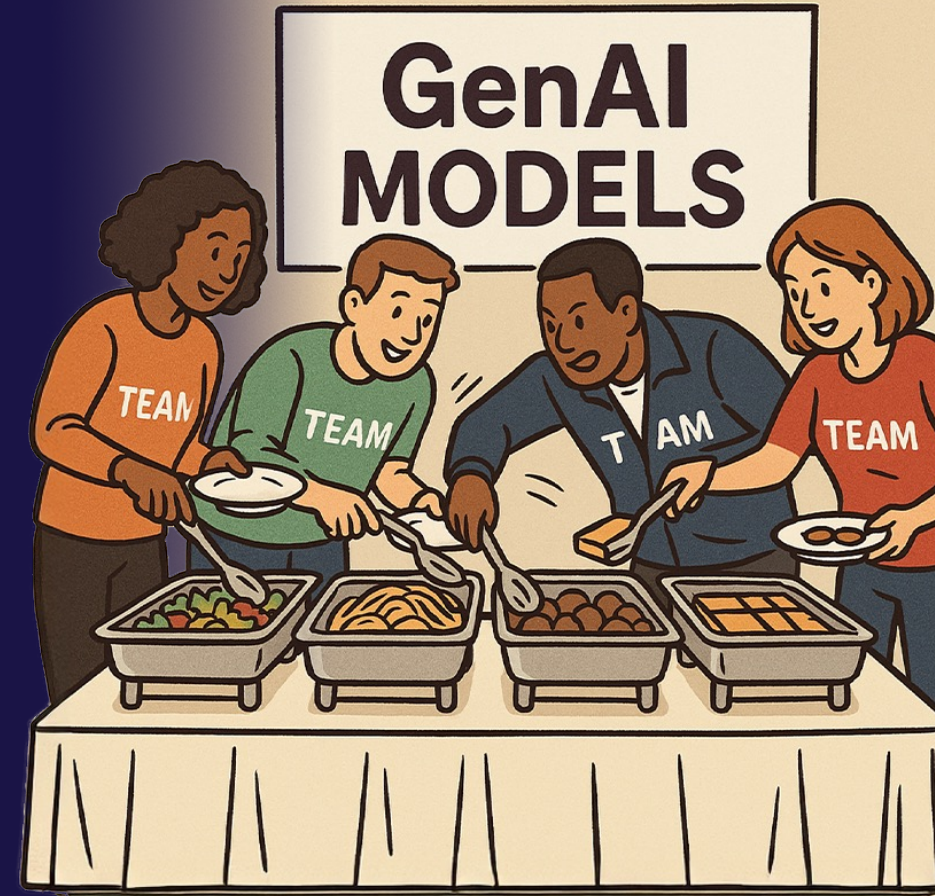
# Goal: Self-service access

## We believe in giving engineers autonomy

Giving teams self-service access to tools & tech  
Trusting them to figure out the best outcomes  
All well before the “platform engineering” age

## We assumed this would work for GenAI

Make the models broadly available  
Let teams experiment and adopt



# Problem: GenAI adoption != Cloud adoption

## Initial access to models was gated

Providers didn't have enough quota

## Concerns were everywhere:

Legal, governance, and security risks

Model quotas and rate limits

Resource allocation conflicts

## Adoption would not be smooth

Potentially fragmented, inefficient & risky



# Problem: The battleship diagram of doom...

## In the past access to models was patchy

They were released where capacity allowed  
In demand models might not be available  
Often the migration path was unclear

## Tokens per minute was the scaling barrier

Sometimes you had to request additional quotas  
We gained quotas by deploying to multiple regions

**Today investment by CSPs in Data Centers has solved most problems in core regions**

| Chat completions             |                           |                        |                       |                       |                       |                            |                            |                       |                       |
|------------------------------|---------------------------|------------------------|-----------------------|-----------------------|-----------------------|----------------------------|----------------------------|-----------------------|-----------------------|
| <a href="#">Expand table</a> |                           |                        |                       |                       |                       |                            |                            |                       |                       |
| Region                       | o1-preview,<br>2024-09-12 | o1-mini,<br>2024-09-12 | gpt-4o,<br>2024-05-13 | gpt-4o,<br>2024-11-20 | gpt-4o,<br>2024-08-06 | gpt-4o-mini,<br>2024-07-18 | gpt-4-turbo,<br>2024-04-09 | gpt-35-turbo,<br>1106 | gpt-35-turbo,<br>0125 |
| australiaeast                | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | ✓                     | ✓                     |
| canadaeast                   | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | ✓                     | ✓                     |
| eastus                       | ✓                         | ✓                      | ✓                     | ✓                     | ✓                     | ✓                          | ✓                          | -                     | ✓                     |
| eastus2                      | ✓                         | ✓                      | ✓                     | ✓                     | ✓                     | ✓                          | ✓                          | -                     | ✓                     |
| francecentral                | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | ✓                     | ✓                     |
| japaneast                    | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | -                     | ✓                     |
| northcentralus               | ✓                         | ✓                      | ✓                     | ✓                     | ✓                     | ✓                          | ✓                          | -                     | ✓                     |
| norwayeast                   | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | -                     | -                     |
| southcentralus               | ✓                         | ✓                      | ✓                     | ✓                     | ✓                     | ✓                          | ✓                          | -                     | ✓                     |
| southindia                   | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | ✓                     | ✓                     |
| swedencentral                | ✓                         | ✓                      | ✓                     | ✓                     | ✓                     | ✓                          | ✓                          | ✓                     | ✓                     |
| switzerlandnorth             | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | -                     | ✓                     |
| uksouth                      | -                         | -                      | -                     | ✓                     | -                     | -                          | -                          | ✓                     | ✓                     |
| westeurope                   | -                         | -                      | -                     | -                     | -                     | -                          | -                          | -                     | ✓                     |
| westus                       | ✓                         | ✓                      | ✓                     | ✓                     | ✓                     | ✓                          | ✓                          | ✓                     | ✓                     |
| westus3                      | ✓                         | ✓                      | ✓                     | ✓                     | ✓                     | ✓                          | ✓                          | -                     | ✓                     |





# Solution: Build an AI platform

## A platform lets us control resources

- Manage access to different models
- Allocate quotas to control usage
- Prevent use-cases interfering

## We have a centralised pain of glass

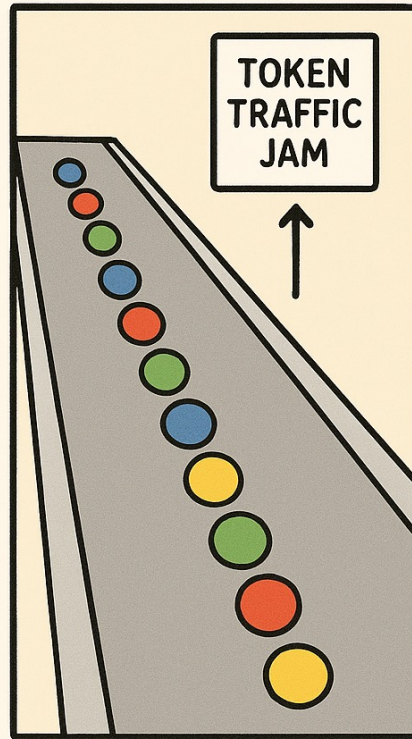
- To apply guardrails and consistency
- To have a clear view into and manage cost

## Eventually move to a self-service approach

- In line with how we manage other resources
- As we automate solutions to day 2 challenges



## SMOOTH SCALING



## REALITY

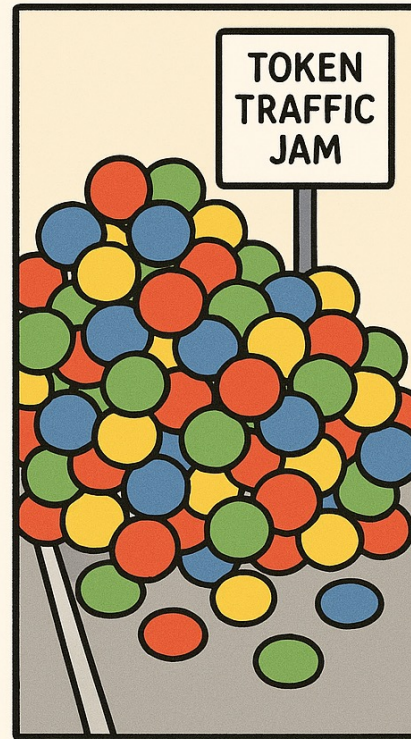


Image generated by Gemini using the prompt: "A side-by-side illustration of two conveyor belts with dots. On the left is smooth scaling where all the dots are moving along without interruption. On the right is a pile up of dots. Both have a sign reading 'Token Traffic Jam' in the background."



# Lessons to learn

**Scaling GenAI needs shared rails**

**A centralised platform**

To enforce guardrails

**Standard APIs**

To simplify access across teams

**Dashboards & cost controls**

For sustainable use

**Autonomy still matters**

But within a safe, governed framework





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Today's fix is  
tomorrow's pain

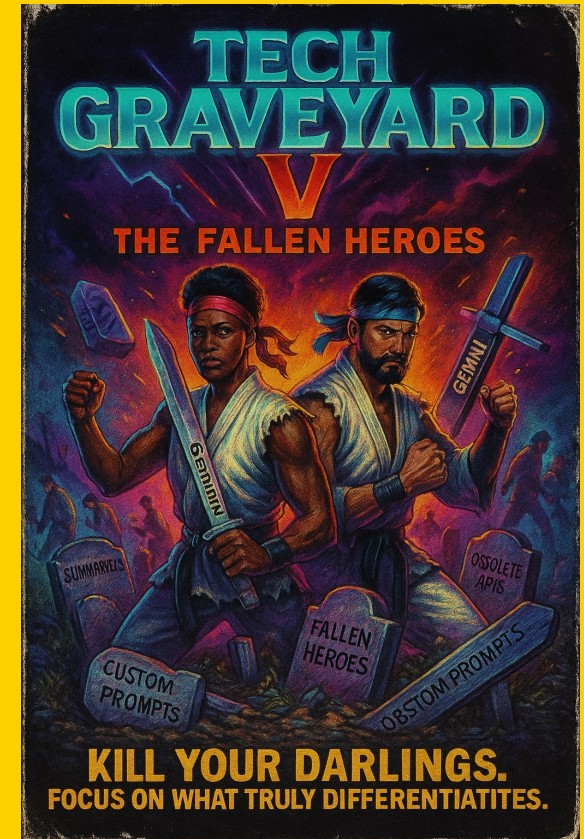


Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is 'Tech Graveyard V: The Fallen Heroes'. Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artist walking through a graveyard of broken weapons/tools labeled Summarizer, Custom Prompts, Obsolete APIs. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: 'Kill your darlings. Focus on what truly differentiates.'"



# Problem: Rapid obsolescence in services

## Half-life of capabilities is very short

If it's deployed, then it's legacy 😊

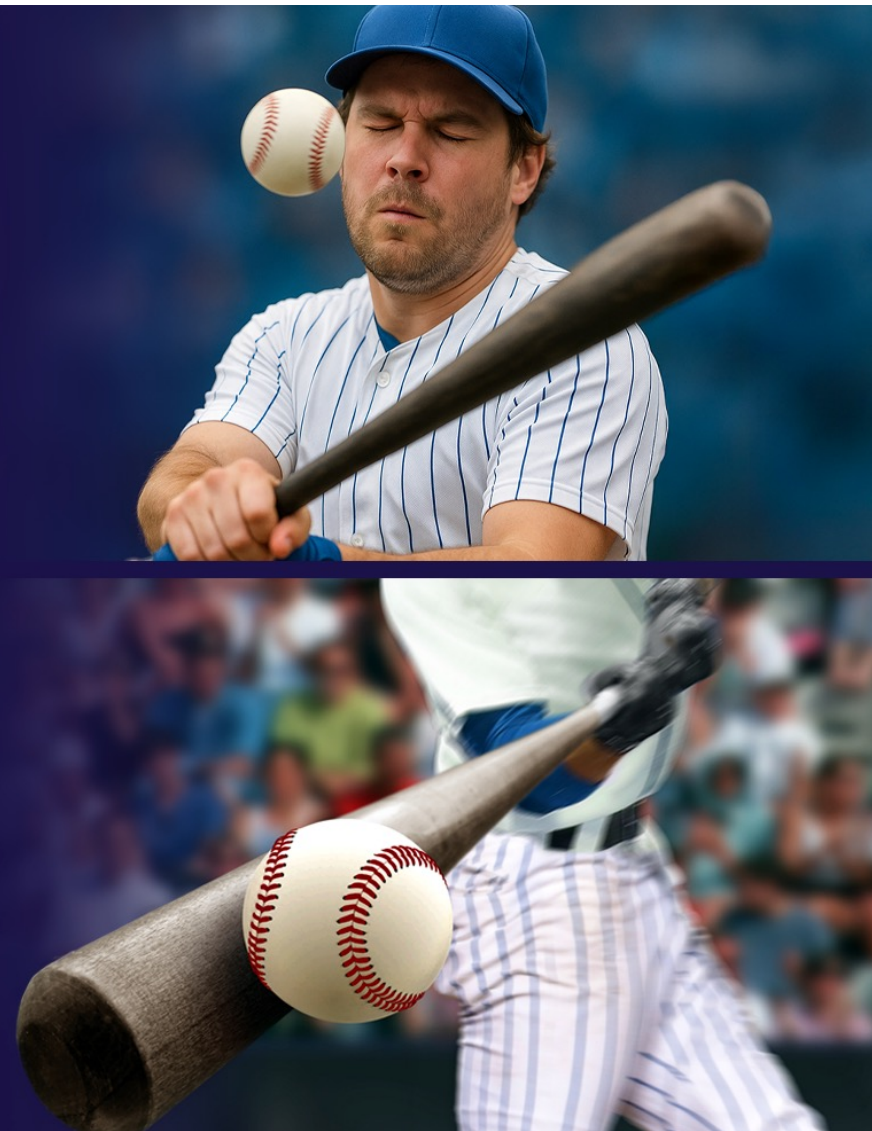
## Consider our Summarising Service

GPT 3.5 has a 4,096 token / ~6 page limit

We created a wrapper / facade to allow larger bodies of text to be summarized

## This is now mostly obsolete

Modern models allow 128k / ~192 pages



# Problem: Rapid obsolescence in methods

## The same issue applies to methods

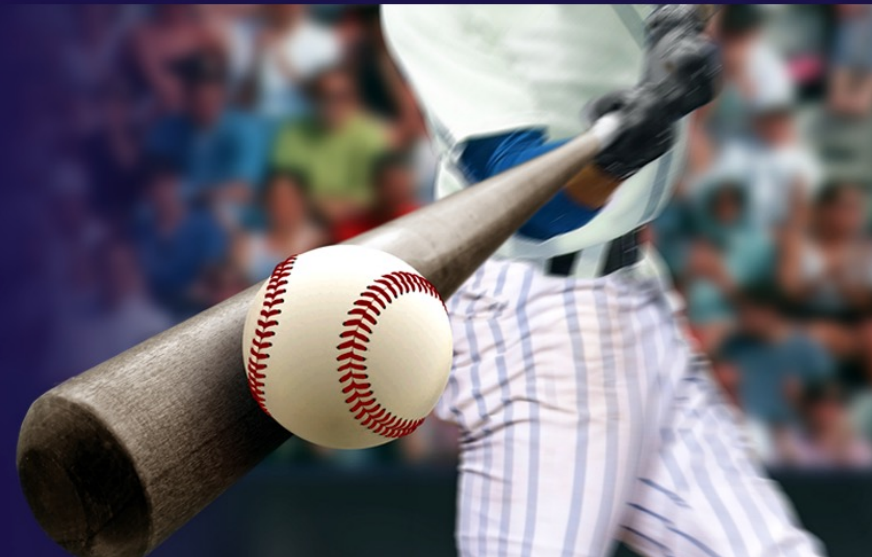
There is no moat to allow percolation time

What was once seen as differentiating is rapidly co-opted by The Big Guys™

## Consider instructions files in CoPilot

We were proving out our own stored prompts / instructions files

But [Awesome GitHub Copilot Customizations](#) is the emerging standard





In essence we need  
a Tech Graveyard to  
bury products being  
outpaced



Image generated by ChatGPT using the prompt: "A graveyard with two tombstones that read 'RIP LM Summarizer' and 'RIP AI Platform'."



# Lessons to learn

**The half life of GenAI solutions is short**

**Adopt a composable architecture**

As always, modularity matters

**Be ready to 'kill your darlings'**

"There's no shame in our game"

**Focus on what *truly differentiates***

What makes your business different?

What can no one else provide?





6

# Stay out of the weeds



Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is "Dojo Drift III: Lost in the Scrolls". Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artists tangled in giant scrolls and blueprints, fighting with compasses and rulers instead of swords. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: "Don't chase perfection. Alignment beats elegance.""





# Problem: Stuck in the weeds

**We wanted a GenAI North Star architecture**

**Instead, we were bogged down:**

- Debates on terms (Assistant vs. Copilot vs. Agent)
- Trying to define our Ubiquitous Language for GenAI
- Iterating over diagrams without making progress

**We needed a reality check**

- Focusing on the map distracted us from the journey
- We risked creating a beautiful designs no one used



# Solution: Move fast and bruise things

## We drew a line in the sand

Decided the perfect is the enemy of the good

## We defined a north star AI architecture which:

Helped us react quickly to emerging tech

Scaled to tens of thousands of regular users

Enabled both alignment and experimentation

Allowed us to deliver significant use cases

## This is now in its second iteration

Bain & Co have benchmarked it against the industry

We can industrialize to support AI use cases at scale



# Lessons to learn

## **Don't chase perfection**

Either in terminology or diagrams  
They will change apace anyway

## **Timebox debate, publish, and iterate**

The value isn't in the architectural diagram  
It's in the shared direction and terminology

## **A viable North Star is all you need**

Infinitely preferable to having no North Star  
Progress over perfection





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# Costs can spiral

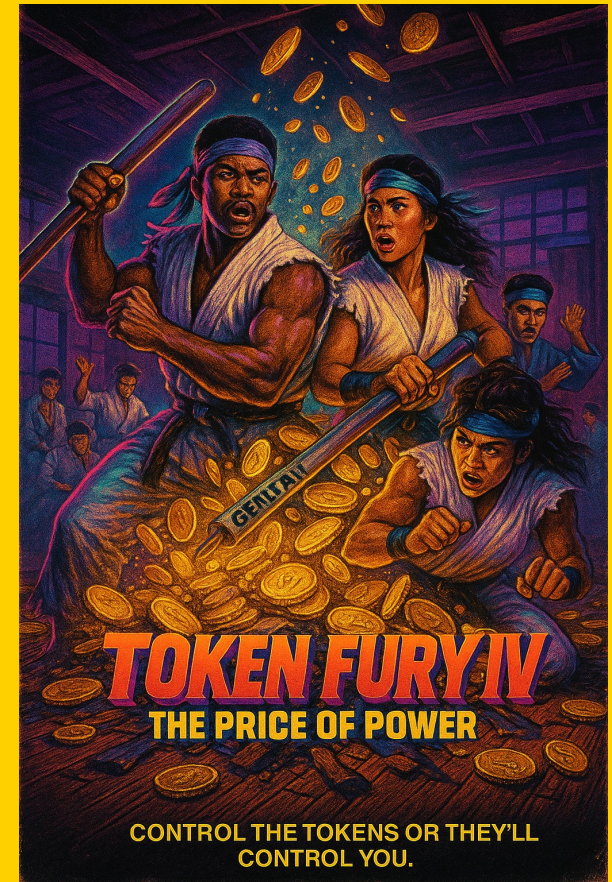


Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is "Token Fury IV: The Price of Power". Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artist swamped by glowing coins/tokens falling from the ceiling, smashing through the dojo floor. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: "Control the tokens or they'll control you."



# Problem: Potential for excessive spending

Here GenAI is the same as Cloud

See [Craig McCarter The myth of scaling serverless systems \(SDB\)](#)

## Good news:

You can process a huge data set via the latest frontier models

## Bad news:

You can process a huge data set via the latest frontier models!!

**“Increasing performance increases the ability of your system to incur cost”**



# Problem: Potential for excessive spending

Costs can change with model version

Consider the GPT family of models:

GPT 4.1 → \$8 per million output tokens

GTP 5 → \$10 per million output tokens

Claude Opus 4.1 → \$75 per million!!!

**Junior engineers can burn \$\$\$**

With one bad prompt or model choice

We don't want devs stopping for the day when the token limit is exhausted





# Building shouldn't be like Barry's



Source: Wikimedia Commons (licensed under CC BY-SA 2.0)



# A Very Bad Day

## **Alerting notified us of significant spend**

“What app is using up all our tokens!!!”

## **A service had been running for some time**

Buring through ~\$20K in the process

“Should we push the kill switch?”

## **We tracked down use case owner**

They needed to process a large corpus of data

The use case was completely legitimate

## **Switching to a different model saved \$40k**



# Solution: Take control!

## Build Cost Tracking Middleware

That tracks spend in real time

## Manage quotas

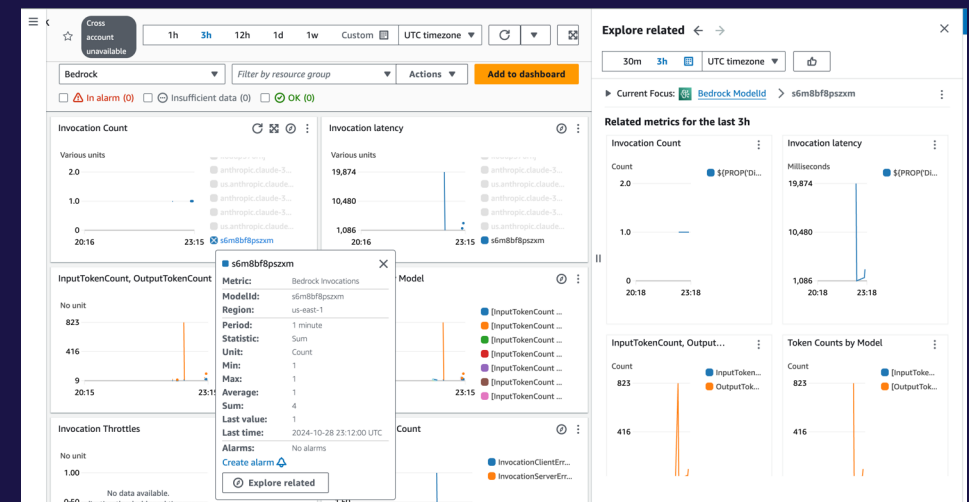
At the team or project to prevent cap usage

## Provide dashboards

To monitor usage and spend by team/project

## Automate alerting and Notifications

Trigger during spikes or near cost thresholds



(A standard Amazon Bedrock dashboard)

Source: <https://aws.amazon.com/blogs/machine-learning/track-allocate-and-manage-your-generative-ai-cost-and-usage-with-amazon-bedrock/>





# Lessons to learn

**Give people a choice of models**

**Educate them on cost implications**

**Observability is crucially important**

**Consider old-school options**

Techniques like batch processing were designed to conserve resources



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# RAG is really hard

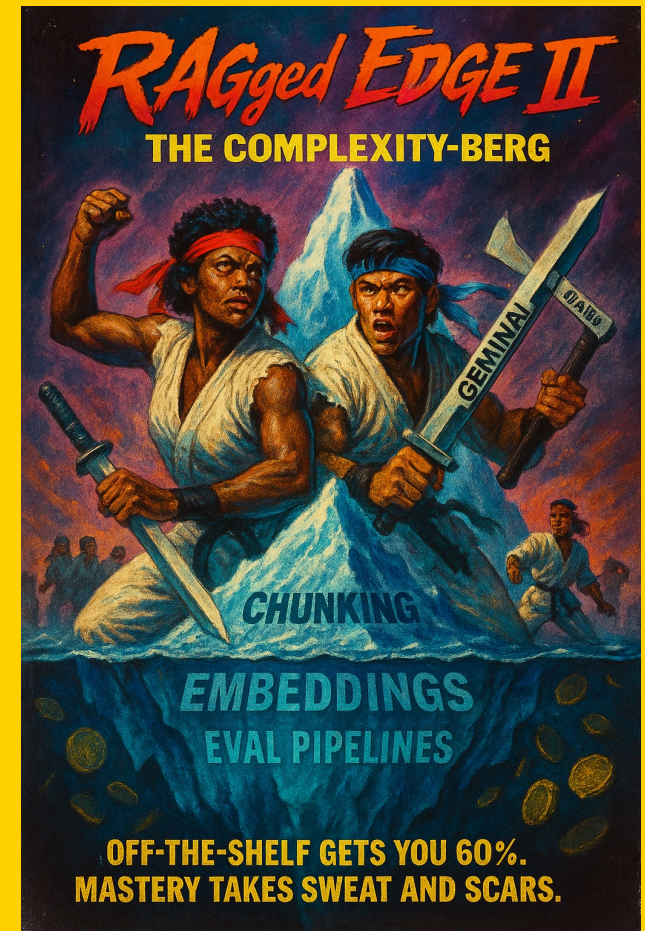


Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is 'Title: RAGged Edge II: The Complexity-Berg'. Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artists climbing a giant iceberg where only the tip is visible, the massive underside labeled Chunking, Embeddings, Eval Pipelines.. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: 'Off-the-shelf gets you 60%. Mastery takes sweat and scars.'"



# Problem: The Complexity-berg of RAG

RAG looks easy in the beginning

Lots of libraries and tutorials

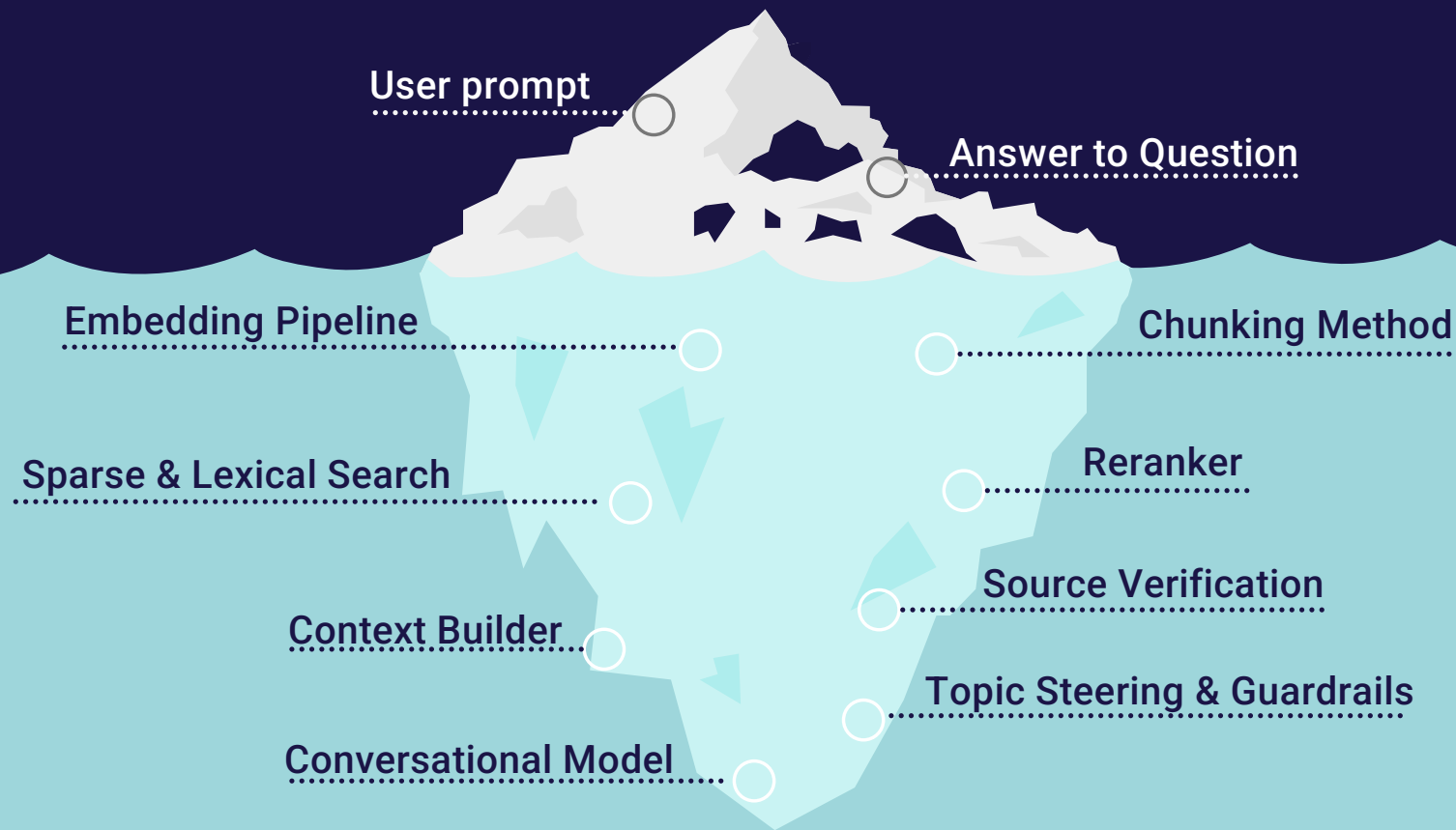
**But the devil is in the details**

Getting the correct response  
...in the format you want  
...with the right level of detail





# Problem: The Complexity-berg of RAG



# Problem: The Complexity-berg of RAG

Off the shelf libraries will take you to 65%

**Going further requires deep knowledge of**

- Embedding and chunking methods

- Searching and reranking process

- Creating effective prompts

**You may have additional needs:**

- Add a conversational component

- Perform tasks such as summarization



# Problem: The Black Box of Non-Determinism

## Generative AI models are generative 😊

Output may not be accurate or trustworthy

The same inputs don't give the same outputs

Outputs are always given with confidence

## There are remedies

Grounded-ness: indicates information in the output that was not in the input

Citations: provides a link to the source material that was used to generate the output





# Solution: We self-educated

## Learn how to answer the crucial questions:

- What Model / Vector Store / Chunking Strategy / Embedding Model should we use?
- How will we evaluate our answers?
- Do we need custom evaluation metrics?
- How can I build trust with the users?
- How will we manage citations?
- How will we manage grounded-ness?
- Must we extract data from images, charts, etc.?

## You will need:

An automated evaluation pipeline

A golden dataset of questions and answers

**New papers and libraries  
will drop daily – so keep up!**



# Lessons to Learn

## It's more than libraries & embeddings

Tutorials will only get you 60–70% of the way

## Precision matters

The right response at the right level of detail is harder than retrieving “something relevant.”

## Expect non-determinism

Tune your pipelines but accept unpredictability

## Grounding is essential

Citations, trust, and verification are not optional if you want production-grade systems

## RAG ≠ one-and-done

Continuous tuning is needed to maintain quality  
Index updates, prompt tuning, evals etc.



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# Latency is real



Image generated by ChatGPT using the prompt: "Create a retro 1980s martial arts action movie poster. The title is "Fast Fury V: The Waiting Game". Show two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). They are in dramatic poses, wearing vintage martial arts uniforms with 80s flair (bandanas, ripped sleeves, neon glow). The background should exaggerate the metaphor: Martial artists mid-fight, but their punches land in slow motion while opponents dodge easily. Use bold, distressed typography, vibrant neon colors, and a gritty VHS-style texture to make it feel like a classic 80s action sequel. Add a dramatic tagline at the bottom: "Speed matters. Optimize or be left behind.""





# Problem: AI based features can lag

## Response times from models can be slow

So chaining API calls will make solutions slow

## Initially we focused on use cases

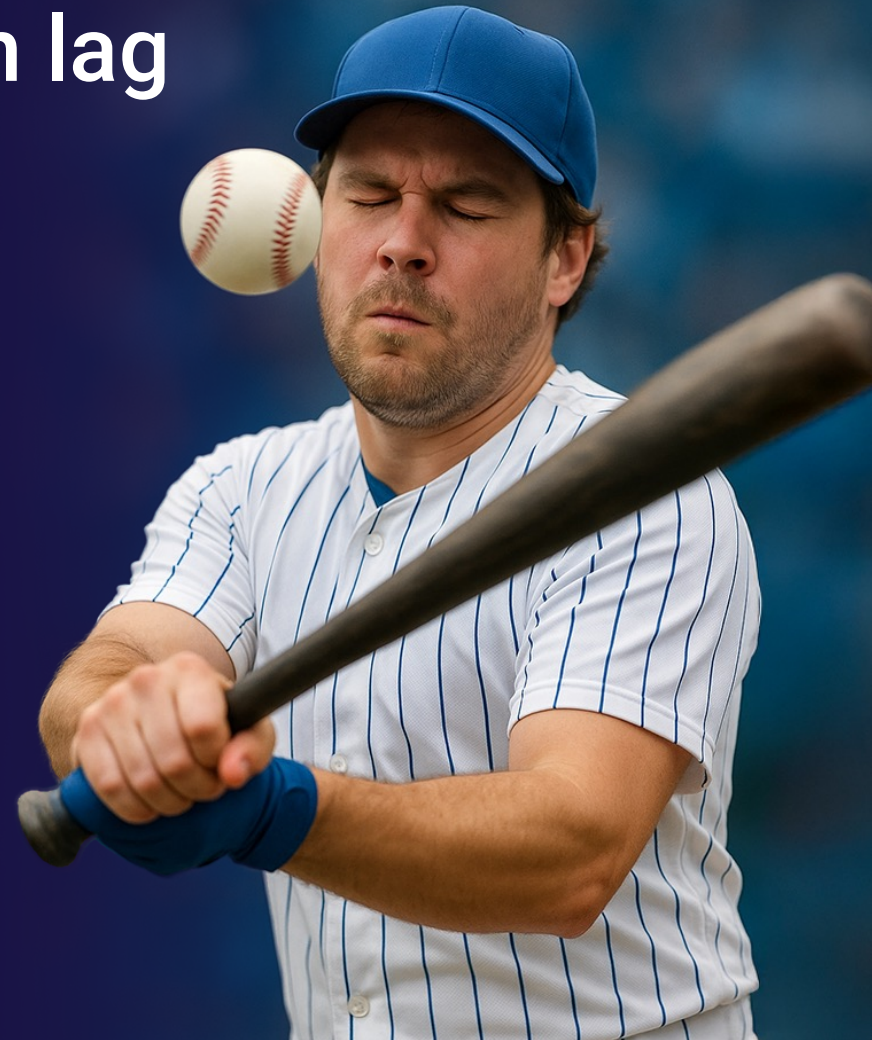
Specifically, the functional requirements

Hoping users would wait for quality outputs

But they found 15 second delays frustrating

## Users leant back into familiar tools

Adoption slowed and confidence dropped



# Solution: Optimize the use of AI

## We can optimize AI

Just like we would any other system

## In the previous case we:

Introduced GPU compute for a 6-fold speed up

Re-indexed and optimised the Vector Database

Fine tuned the embeddings model

Took advantage of DB upgrades

Redesigned the UX to keep the user engaged

**Response time fell from 15 to 3 seconds**



# Lessons to Learn

## Perception beats absolute speed

Users care about feeling the system is responsive  
Progress updates and clever UX go a long way

## Use the right model for the job

Go smaller & faster where precision isn't critical.

## Design for trade-offs

Latency, cost, accuracy are the 'dials you can turn'  
Pick the setting that work best for your system

## Measure latency based on context

A chatbot for Q&A can tolerate 3–4 secs  
In customer support under secs 2s is key





# Summary



Create a retro 1980s video store shelf featuring nine VHS movie covers. Each cover is styled as a martial arts action sequel, with bold neon colors, distressed textures, and over-the-top titles, showing two diverse martial arts heroes who are software architects (one woman, one man, different ethnicities, stylized, no resemblance to real people). The movies are: Endless Dojo IV: The Never-Ending Lesson Adoption Story III: The Reluctant Students Model Warriors II: Clash of the Titans Scaling is a Beast II: The Reckoning Tech Graveyard V: The Fallen Heroes Dojo Drift III: Lost in the Scrolls Token Fury IV: The Price of Power RAGged Edge II: The Complexity-Berg Fast Fury V: The Waiting Game Make it look like a classic 80s VHS rental wall, with colorful covers lined up side by side. Add a gritty VHS texture, glowing neon accents, and dramatic martial arts artwork on each box. The overall vibe should be playful, nostalgic, and over-the-top.



# What have we learned?

1. Learning never stops
2. Shipping isn't adoption
3. Models aren't equal
4. Scaling is a beast
5. Today's fix is tomorrow's pain
6. Stay out of the weeds
7. Costs can spiral
8. RAG is really hard
9. Latency is real



# Go Forth and Skin Your Knees



**Swing & Miss:** Things rarely work the first time

**Reality Check:** Scaling AI is messy, chaotic, and full of surprises

**Lesson Learned:** Mistakes are signals — if you learn fast, you win







**Liberty™**  
**Information**  
**Technology**

**goto;**

# **QUESTIONS?**

ASK ME IN THE APP

goto;

# **RATE THIS SESSION**

YES, YOU'LL GET A REWARD